Bhanu Kiran R

9686314859 | bhanukiran90216@gmail.com
linkedin.com/in/bhanu-kiran-r | github.com/Bhanucreator | bhanu-kiran.vercel.app

Education

C Byregowda Institute of Technology — Kolar, Karnataka
Bachelor of Engineering in Computer Science — CGPA: 8.1 — *Aug 2023 – May 2027*

- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Cloud Computing, Software Engineering.
- Vice President, Club of Programmers (COPS CBIT) — leading technical programs for 200+ student members.

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL, C, HTML, CSS
Frameworks: React.js, React Native, FastAPI, Flask, Node.js
Databases: PostgreSQL, MySQL, MongoDB, PostGIS, Redis
DevOps and Tools: AWS, Docker, Kubernetes, Git, GitHub Actions, CI/CD Pipelines, Linux
Architecture: REST APIs, WebSockets, Microservices

Experience

Software Engineering Intern — Jan 2026 – Feb 2026
Microsoft x AICTE — *Remote*

- Completed a 4-week intensive program on Microsoft emerging technologies covering software engineering principles, cloud computing, and system design.
- Accumulated 85+ hours of hands-on training in cloud architecture, Azure services, and scalable application development.
- Applied software development best practices including modular design, version control with Git, and deployment on cloud infrastructure.

Vice President — Aug 2023 – Present
Club of Programmers (COPS), CBIT Kolar — cops-official.vercel.app — *Kolar, Karnataka*

- Progressed from member to Vice President; planned and executed coding competitions, technical workshops, and i-Sphere 1.0, an inter-collegiate techno fest with 200+ attendees.
- Led full-stack development of the official club website end-to-end alongside the Technical Lead, centralising resources and improving online presence.
- Mentored 15+ junior developers in full-stack development, data structures, and algorithms, achieving an 80% project completion rate.

Projects

VIKRANTA – Real-Time Tourist Safety Platform | *React, Flask, PostgreSQL, PostGIS, Docker, WebSockets* | Live Demo — Oct 2025

- Designed and built a full-stack real-time safety platform using a microservices architecture, enabling live distress signalling and geofenced alerts for tourists.
- Engineered a WebSocket-based event-driven communication layer handling 50+ concurrent connections with a 98% alert delivery rate and sub-3-second notification latency.
- Optimised geospatial query performance by 85%, achieving sub-100ms response times using PostGIS spatial indexing across 10,000+ indexed danger zones.
- Containerised all services with Docker and deployed on Railway with horizontal auto-scaling, sustaining 200+ concurrent active user sessions under peak load.

IPC Legal Assistant – Cross-Platform Mobile Application | *React Native, Flask, LangChain, REST API* | GitHub — Jan 2025

- Built a production-grade cross-platform mobile application (iOS and Android) using React Native (Expo) with a Python Flask REST API backend deployed on Render.
- Implemented a hybrid keyword-matching and semantic search engine over a 227-page legal corpus, achieving 92% retrieval accuracy and sub-800ms query response times.
- Automated the full CI/CD pipeline using GitHub and Render; backend operates on 512MB RAM with 99.5% uptime across a 60-day production period.

Python AI Tutor – Asynchronous Chat Platform | *FastAPI, WebSockets, Redis, Docker, GitHub Actions* | GitHub — Dec 2024

- Developed an asynchronous, concurrent web application using FastAPI and WebSockets handling 200+ simultaneous connections with 99.8% uptime over a 90-day production cycle.
- Implemented Redis-based session management for distributed context pooling, reducing average API response latency to 450ms.
- Automated containerised deployment using GitHub Actions CI/CD pipelines, cutting release cycle time by 75%.

ImageClassifier – CNN Image Recognition System | *Python, TensorFlow, Keras, ResNet50* | Live Demo — Nov 2025

- Architected and trained a multi-class CNN using transfer learning with a pre-trained ResNet50 backbone, achieving 94.3% top-1 accuracy on a 50,000+ image dataset.
- Applied model quantisation to reduce deployment artifact size by 75% (98MB to 24MB), cutting CPU inference latency from 145ms to 38ms for edge deployment readiness.

Publications

KisanShakti: Integrated Smart Farming and Hyperlocal Commerce Platform | ICNEXT-2026 — 2026

- Peer-reviewed paper accepted at the International Conference on Next-Gen Engineering and Technology; ISBN: 978-93-5768-920-5.
- Engineered an offline-capable React Native architecture for rural low-connectivity zones, integrating IoT data streams and a marketplace bypassing intermediaries capturing 60–70% of crop value.

AI Applications in Pharmaceutical Drug Discovery | NCRTEST-2025 — 2025

- Published at the 7th National Conference on Recent Trends in Engineering Science and Technology; applied deep learning techniques reducing computational cost by 35% over traditional methods.

Achievements and Certifications

Certificate of Appreciation – Hack Fest ’25 | Bangalore Technological Institute — Apr 2025

- Ranked in the top 5 teams out of 40+ competing groups for designing an innovative solution to a real-world engineering challenge.

Machine Learning, Deep Learning and Generative AI | DHS Informatics — Oct 2025

- Completed 40 hours of structured training in supervised and unsupervised learning, neural network architectures, and large language model applications.

Computer Networks using Cisco Packet Tracer | Cisco — Oct 2025

- Practical training in network topology design, routing protocols (OSPF, BGP), and network troubleshooting using Cisco tools.